

SmolVLM for Dense Video Captioning

Week 10: Internship Update

VideoMME

- 900 video, 3 multiple-choice questions each
- 6 visual domains: Knowledge, Film & Television, Sports Competition, Life Record, and Multilingual
- 30 video categories in total
- 12 task categories including perception, reasoning and information synthesis
- All videos between 11 seconds and 1 hour
- Short videos (< 2 minutes), medium videos (4-15 minutes) and long videos (30-60 minutes)

Ovis2-2B

- Different frame sampling techniques

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Short	36.8%	35.5%	36.3%	36.1%	35.9%
Medium	35.4%	35.6%	36.7%	35.0%	36.0%
Long	35.2%	34.5%	34.7%	35.0%	34.2%
Entire dataset	35.8%	35.2%	35.9%	35.4%	35.4%

Table 6: VideoMME results over frame sampling techniques: Model: Ovis2-2B

Ovis2-2B

- Video categories: center (1171.50 s) and first (1085.51 s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	37.0%	37.3%	36.5%	33.3%	35.8%	16.7%	35.5%
Medium	36.6%	40.8%	36.1%	35.7%	29.9%	41.4%	35.6%
Long	32.7%	37.5%	39.5%	36.1%	29.9%	39.3%	34.5%
Entire dataset	35.5%	38.5%	37.3%	35.0%	31.9%	32.2%	35.2%

Table 1: VideoMME results over video categories: Model: Ovis2-2B, Frame sampling technique: center

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	40.2%	32.5%	35.3%	37.5%	37.4%	23.3%	36.8%
Medium	33.7%	38.3%	37.6%	41.4%	29.6%	43.3%	35.4%
Long	35.2%	32.5%	39.6%	39.5%	30.0%	41.4%	35.2%
Entire dataset	36.4%	34.5%	37.5%	39.4%	32.4%	36.0%	35.8%

Table 2: VideoMME results over video categories: Model: Ovis2-2B, Frame sampling technique: first

Ovis2-2B

- Video categories: maxinfo:1000:96 (69741.69 s) and csta:1000:96 (84912.0s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	35.9%	33.3%	38.0%	39.2%	36.2%	26.7%	36.1%
Medium	37.4%	35.8%	34.7%	38.3%	28.6%	43.3%	35.0%
Long	34.8%	35.8%	37.3%	40.0%	29.5%	40.0%	35.0%
Entire dataset	36.0%	35.0%	36.7%	39.2%	31.4%	36.7%	35.4%

Table 3: VideoMME results over video categories: Model: Ovis2-2B, Frame sampling technique: max-info:1000:96

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	38.1%	33.3%	36.0%	38.3%	33.3%	33.3%	35.9%
Medium	37.8%	38.3%	32.7%	39.2%	32.4%	40.0%	36.0%
Long	36.3%	30.8%	36.7%	36.7%	29.5%	40.0%	34.2%
Entire dataset	37.4%	34.2%	35.1%	38.1%	31.7%	37.8%	35.4%

Table 4: VideoMME results over video categories: Model: Ovis, Frame sampling technique: csta:1000:96

Ovis2-2B

- Video categories: fps:1:4:96 (23632.65 s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	37.4%	30.8%	36.0%	39.2%	36.2%	40.0%	36.3%
Medium	37.4%	40.8%	36.7%	41.7%	29.5%	43.3%	36.7%
Long	36.3%	34.2%	39.3%	36.7%	27.1%	43.3%	34.7%
Entire dataset	37.0%	35.3%	37.3%	39.2%	31.0%	42.2%	35.9%

Table 5: VideoMME results over video categories: Model: Ovis, Frame sampling technique: fps:1:4:96

Ovis2-2B

Video Type	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Temporal Perception	32.7%	34.5%	38.2%	32.7%	40.0%
Spatial Perception	35.2%	37.7%	37.0%	37.0%	33.3%
Attribute Perception	38.2%	40.9%	40.1%	38.3%	40.1%
Action Recognition	38.1%	36.2%	38.0%	37.7%	38.0%
Object Recognition	31.1%	31.1%	32.2%	32.5%	30.5%
OCR Problems	35.3%	36.0%	36.7%	34.5%	32.4%
Counting Problem	24.2%	25.0%	30.2%	26.9%	28.0%
Temporal Reasoning	32.6%	34.0%	32.8%	37.3%	31.6%
Spatial Reasoning	37.7%	44.6%	32.1%	32.1%	33.9%
Action Reasoning	35.1%	32.3%	34.4%	33.0%	34.4%
Object Reasoning	36.3%	38.1%	37.0%	37.9%	38.1%
Information Synopsis	43.2%	44.8%	40.9%	39.9%	41.2%

Table 7: VideoMME results over task categories: Model: Ovis2-2B

Qwen2.5-VL-3B-Instruct

- Video categories: center (1294.86 s) and first (1255.91 s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	35.9%	32.5%	35.3%	35.8%	33.8%	43.3%	35.1%
Medium	35.6%	36.7%	30.7%	31.7%	29.5%	36.7%	33.0%
Long	37.0%	35.0%	33.3%	38.3%	25.2%	40.0%	33.7%
Entire dataset	36.2%	34.7%	33.1%	35.3%	29.5%	40.0%	33.9%

Table 8: VideoMME results over video categories: Model: Qwen2.5-VL-3B-Instruct, Frame sampling technique: center

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	40.4%	37.5%	36.7%	40.8%	35.7%	33.3%	38.1%
Medium	38.5%	37.5%	26.0%	33.3%	31.4%	43.3%	34.1%
Long	40.0%	37.5%	36.7%	39.2%	26.7%	36.7%	35.8%
Entire dataset	39.6%	37.5%	33.1%	37.8%	31.3%	37.8%	36.0%

Table 9: VideoMME results over video categories: Model: Qwen2.5-VL-3B-Instruct, Frame sampling technique: first

InternVL3-2B

- Video categories: center (1354.57 s) and first (1284.10 s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	38.9%	40.0%	38.7%	35.8%	35.2%	30.0%	37.4%
Medium	34.1%	43.3%	29.3%	37.5%	30.0%	43.3%	34.3%
Long	35.6%	39.2%	36.7%	35.0%	27.1%	23.3%	33.8%
Entire dataset	36.2%	40.8%	34.9%	36.1%	30.8%	32.2%	35.2%

Table 11: VideoMME results over video categories: Model: InternVL3-2B, Frame sampling technique: center

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	36.7%	35.8%	32.7%	40.8%	32.4%	20.0%	34.9%
Medium	35.2%	40.0%	28.7%	30.8%	29.5%	30.0%	32.7%
Long	36.3%	37.5%	32.0%	33.3%	28.1%	33.3%	33.3%
Entire dataset	36.0%	37.8%	31.1%	35.0%	30.0%	27.8%	33.6%

Table 12: VideoMME results over video categories: Model: InternVL3-2B, Frame sampling technique: first

SmolVLM2-2.2B-Instruct

- Video categories: center (1508.23 s) and first (1418.98 s)

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	29.6%	35.8%	28.7%	34.2%	28.6%	33.3%	30.8%
Medium	40.7%	35.8%	33.3%	36.7%	31.0%	36.7%	35.9%
Long	30.7%	32.5%	32.7%	28.3%	30.0%	33.3%	30.9%
Entire dataset	33.7%	34.7%	31.6%	33.1%	29.8%	34.4%	32.5%

Table 14: VideoMME results over video categories: Model: SmolVLM2-2.2B-Instruct, Frame sampling technique: center

Video Type	Knowledge	Film & Television	Sports Competition	Artistic Performance	Life Record	Multilingual	Overall
Short	30.0%	36.7%	28.0%	39.2%	30.0%	43.3%	32.2%
Medium	38.1%	33.3%	32.0%	38.3%	27.6%	40.0%	34.1%
Long	28.9%	30.0%	29.3%	27.5%	31.4%	30.0%	29.6%
Entire dataset	32.3%	33.3%	29.8%	35.0%	29.7%	37.8%	32.0%

Table 15: VideoMME results over video categories: Model: SmolVLM2-2.2B-Instruct, Frame sampling technique: first

Overall comparison so far

Model	first	center	fps:fps:min:max	maxinfo:input:max	csta:input:max
Ovis2-2B	35.8%	35.2%	35.9%	35.4%	35.4%
InternVL3-2B	33.6%	35.2%	%	%	%
Qwen2-VL-2B-Instruct	%	%	%	%	%
Qwen2.5-VL-3B-Instruct	36.0%	33.9%	%	%	%
SmolVLM2-2.2B-Instruct	32.0%	32.5%	%	%	%

Table 17: VideoMME results over frame sampling techniques for different models

Parameters to be changed and tested

- Max number of frames (for every method)
- FPS
 - number of frames per second
- MaxInfo
 - max number of input frames !!!
 - Rank = 8
 - tol = 1.0
 - normalize_svd_output='l2'
- CSTA
 - max number of input frames !!!
 - now scoring every frame -> coils be tested with splitting
 - extracted length of the video